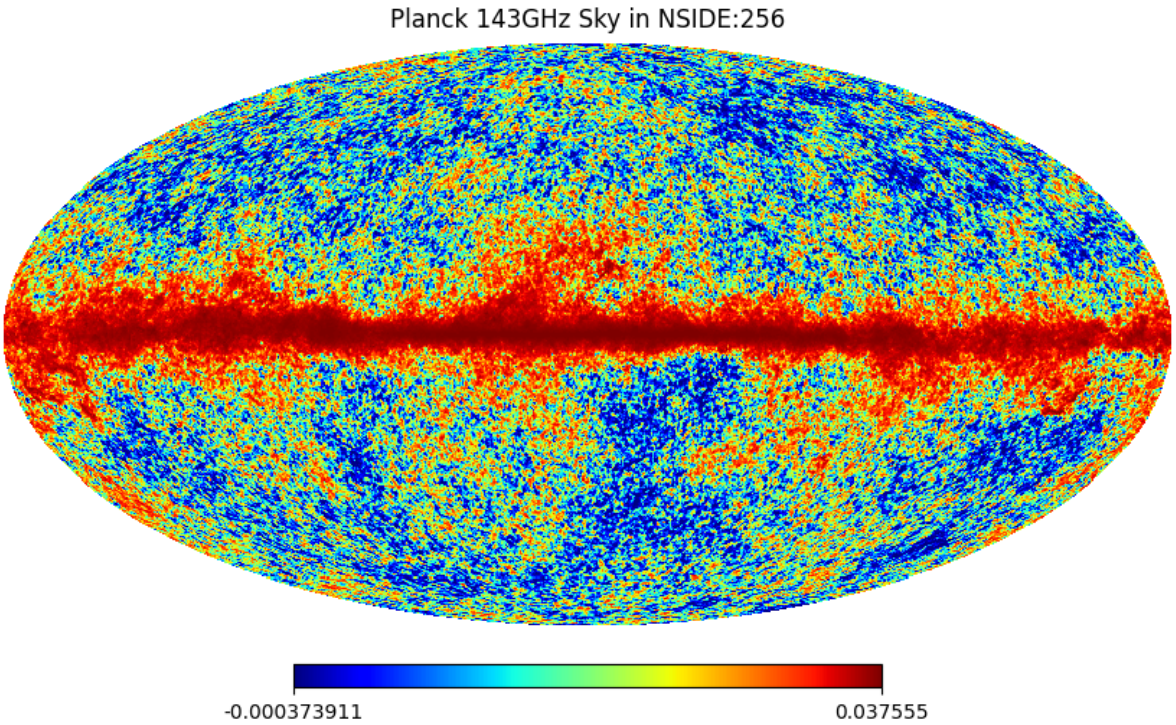
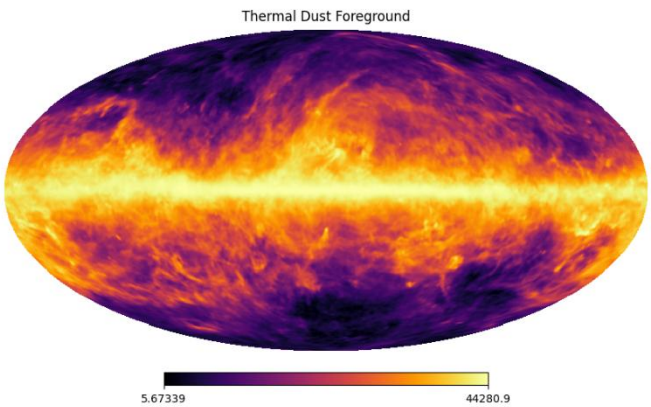
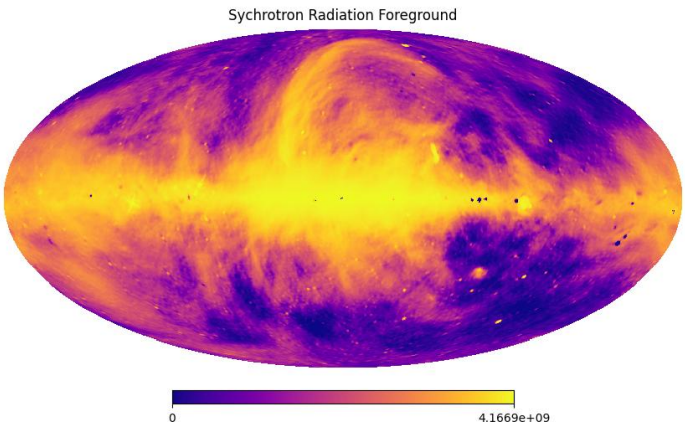
CMB: Foreground Analysis

1. This is the data from the Planck satellite which hasn't been accounted for any noise. This contains foregrounds from Dust, synchrotron, X-foregrounds, etc.



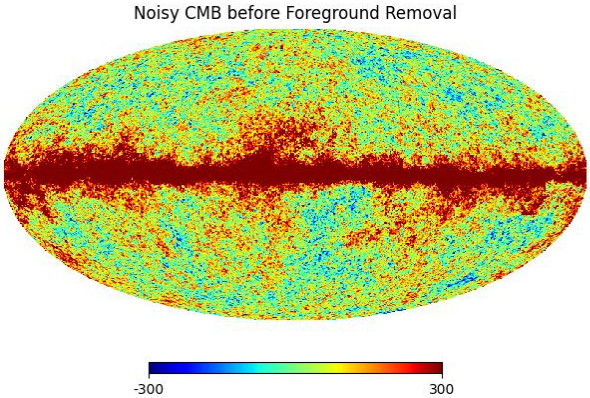
2. We are majorly going to work on synchrotron and dust-dust emissions.

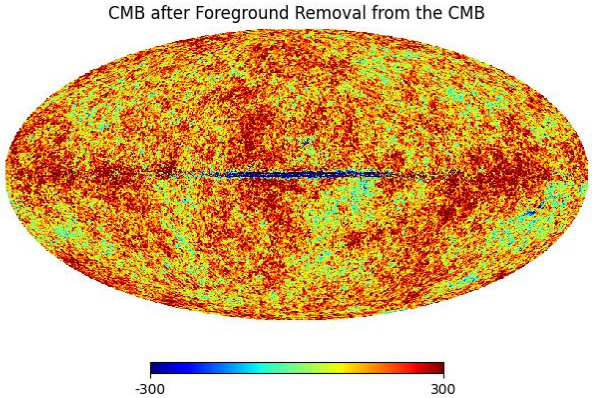




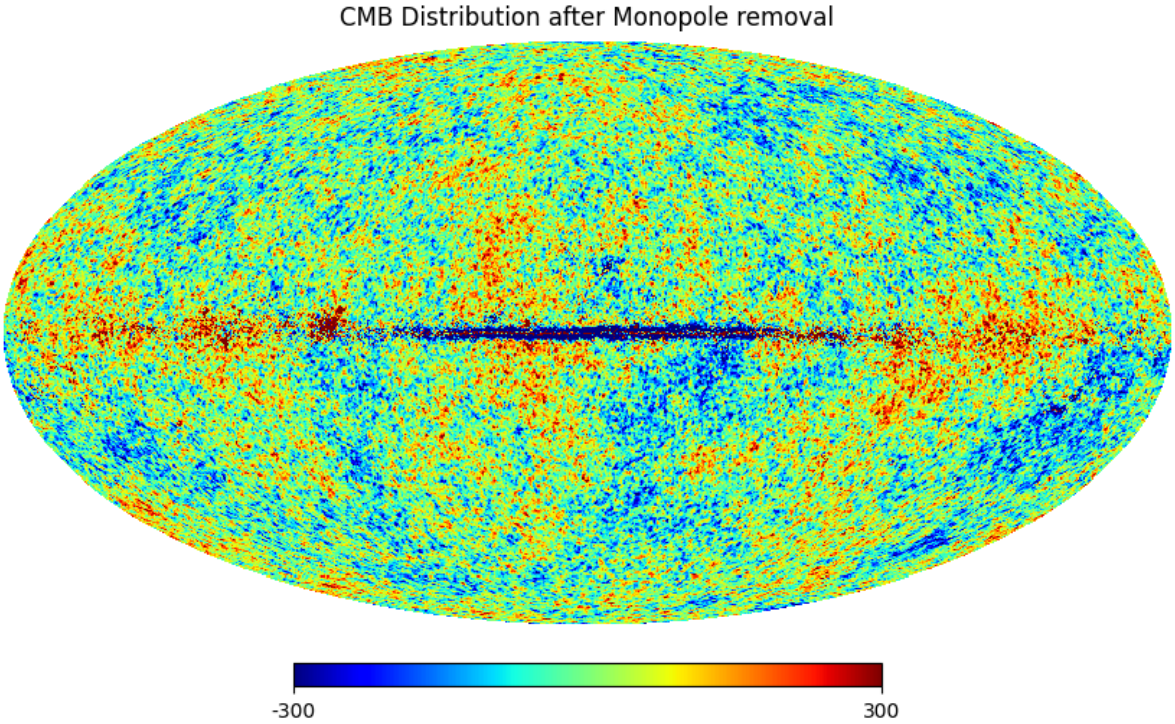
3. Now we downgrade the CMB map from NSIDE 2048 to 256. This ensures the mapping is done one on one when mapping the foreground maps to noisy CMB data.

4. We remove the noise of the synchrotron and dust emissions. We see that the milky way is removed majorly.

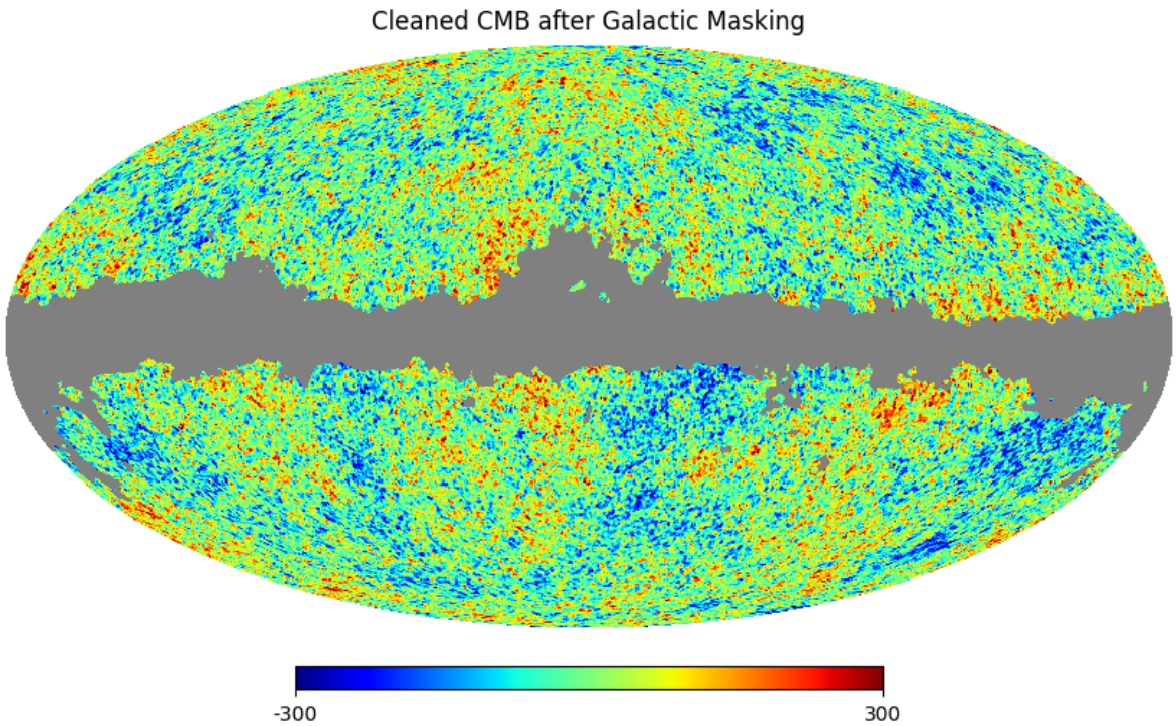




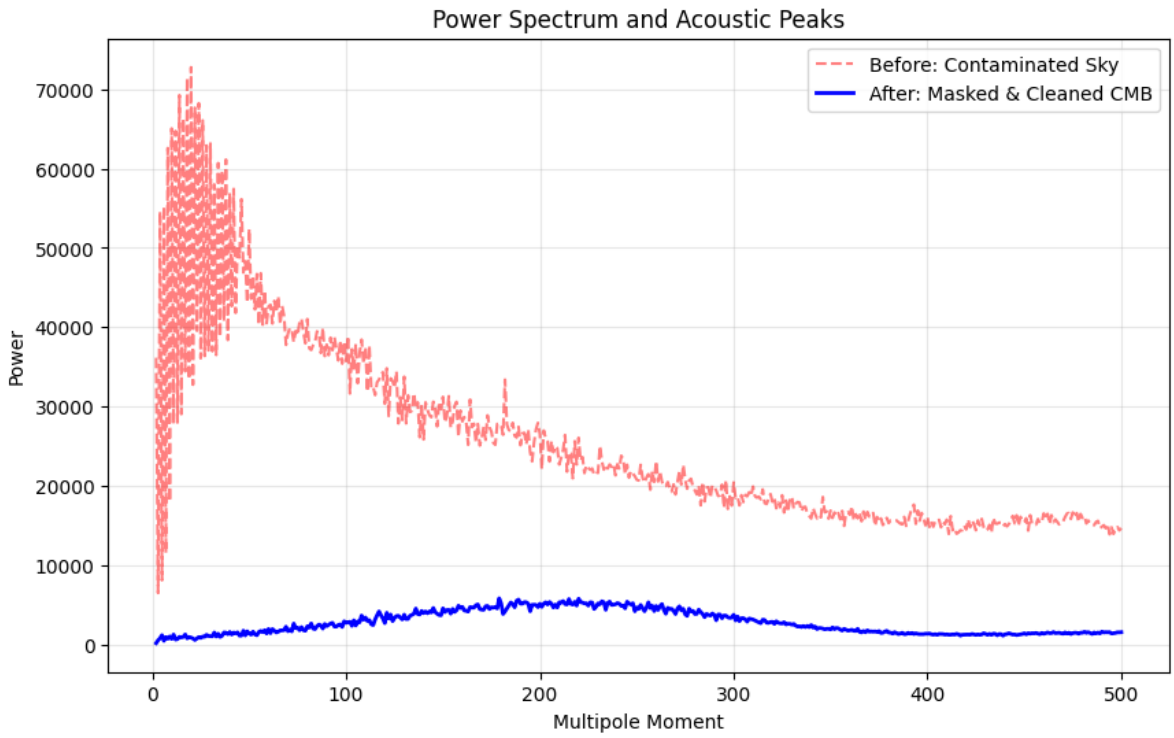
5. We then perform the Monopole Removal method in order to centre the temperature fluctuations centred at zero. We get the below graph.

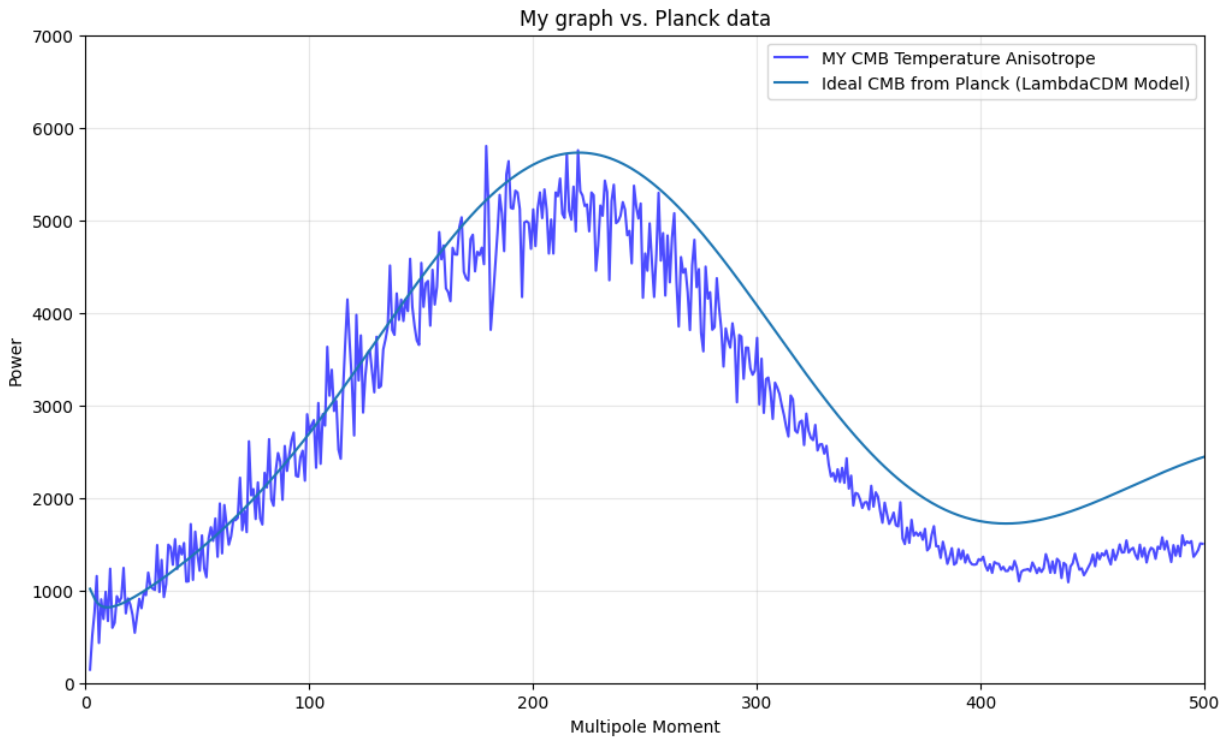


6. To remove any further noise as we can still see is by Galactic Masking.



7. We can draw a power spectrum of this and then compare it with the GOLDEN STANDARD CMB power spectrum achieved by Lambda CDM model. We can compare the two.




My graph vs. Planck data
MY CMB Temperature Anisotrope
Ideal CMB from Planck (LambdaCDM Model)
Power
Multipole Moment
0
1000
2000
3000
4000
5000
6000
7000
0
100
200
300
400
500